



GOVERNMENT OF KARNATAKA
KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD
6TH CROSS, MALLESHWARAM, BENGALURU - 560003

2025-26 II PUC MODEL QUESTION PAPER – 2

SUBJECT: PHYSICS (33)

MAXIMUM MARKS: 70

TIME: 3 HOURS

NUMBER OF QUESTIONS: 45

General Instructions:

1. All parts (A TO D) are compulsory.
2. For Part – A questions, only first written-answer will be considered for evaluation.
3. Answers without relevant diagram / figure / circuit wherever necessary will not carry any marks.
4. Direct answers to numerical problems without relevant formula and detailed solutions will not carry any marks.

PART – A

I. Pick the correct option among the given options for ALL of the following questions: 1×15= 15

1. If a body contains n_1 protons and n_2 electrons, then the net charge of the body is:

- (a) $(n_1 + n_2)e$ (b) $(n_1 - n_2)e$ (c) $(n_2 - n_1)e$ (d) $(n_1 n_2)e$

2. The electric potential at a point, due to a point charge varies with distance from it as:

- (a) $\frac{1}{\text{distance}}$ (b) $\frac{1}{(\text{distance})^2}$ (c) $\frac{1}{(\text{distance})^3}$ (d) $\frac{1}{\sqrt{\text{distance}}}$

3. Kirchoff's junction rule is based on the conservation of:

- (a) mass (b) momentum (c) energy (d) charge

4. The path traced by a charged particle moving perpendicular to a uniform magnetic field is:

- (a) circle (b) straight line (c) helix (d) ellipse

5. Based on the magnetic properties, materials are listed in column – I and values of magnetic susceptibilities are listed in column – II. Identify the correct match.

Column - I	Column - II
(i) Diamagnetic	(p) small and positive
(ii) Paramagnetic	(q) large and positive
(iii) Ferromagnetic	(r) small and negative

- (a) (i) → (q); (ii) → (r); (iii) → (p) (b) (i) → (r); (ii) → (p); (iii) → (q)
(c) (i) → (p); (ii) → (q); (iii) → (r) (d) (i) → (p); (ii) → (r); (iii) → (q)

6. A straight conductor of length ' l ' is moving with a velocity ' v ' in the direction of uniform magnetic field ' B '. The magnitude of induced emf across the ends of the conductor is

- (a) B/v (b) $\frac{B/v}{2}$ (c) 0 (d) $2Blv$

7. When alternating voltage is applied to a resistor, which of the following statements is wrong?
- (a) The average current over a complete cycle is zero.
 - (b) The average voltage over a complete cycle is zero.
 - (c) The average power dissipated by the resistor over a complete cycle is zero.
 - (d) The phase difference between the voltage and current is zero.

8. Which of the following electromagnetic waves are used to kill germs in water purifiers?
- (a) Microwaves (b) Infrared waves (c) Ultraviolet rays (d) Gamma rays

9. The angle of minimum deviation of a prism depends on

- (i) refractive index of the material of the prism
- (ii) refractive index of surrounding medium
- (iii) refracting angle of the prism

- (a) only option (i)
- (b) only option (ii)
- (c) only option (iii)
- (d) all (i), (ii) and (iii)

10. Image of a real object formed by a simple microscope when the object is placed at a distance less than the one focal length is

- (a) real, inverted and magnified
- (b) virtual, erect and magnified
- (c) virtual, erect and diminished
- (d) virtual, inverted and magnified

11. Consider the following two statements regarding diffraction of light.

Statement – I: We do not easily encounter diffraction effects of light in everyday observation.

Statement – II: The wavelength of light is much smaller than the dimensions of most of the obstacles.

Choose the correct option from the following:

- (a) Both the statements are correct and statement–II is the correct explanation for statement–I.
- (b) Both the statements are correct and statement–II is not the correct explanation of statement–I.
- (c) Statement–I is correct and statement–II is wrong
- (d) Both the statements are wrong.

12. The process of emission of electrons from a metal surface by applying high electric field is called

- (a) thermionic emission
- (b) photoelectric emission
- (c) field emission
- (d) secondary emission

13. Let K be the kinetic energy, U be the potential energy and E be the total energy of electron revolving around the nucleus in hydrogen atom, then which of the following is correct?

- (a) $K > 0, U > 0, E > 0$
- (b) $K > 0, U < 0, E < 0$
- (c) $K > 0, U > 0, E < 0$
- (d) $K < 0, U < 0, E < 0$

14. Among the following, which set of nuclei are isotones?

- (a) ${}^{12}_6\text{C}, {}^{13}_6\text{C}$
- (b) ${}^2_1\text{H}, {}^3_2\text{He}$
- (c) ${}^{14}_6\text{C}, {}^{14}_7\text{N}$
- (d) ${}^3_1\text{H}, {}^3_2\text{He}$

15. Carbon, silicon and germanium have four valence electrons each. These are characterised by valence and conduction bands separated by energy band gap respectively equal to $(E_g)_C$, $(E_g)_{Si}$ and $(E_g)_{Ge}$. Which of the following comparison of energy band gaps is correct?

(a) $(E_g)_{Si} < (E_g)_{Ge} < (E_g)_C$

(b) $(E_g)_C < (E_g)_{Ge} > (E_g)_{Si}$

(c) $(E_g)_C > (E_g)_{Si} > (E_g)_{Ge}$

(d) $(E_g)_C = (E_g)_{Si} = (E_g)_{Ge}$

II. Fill in the blanks by choosing appropriate answer given in the bracket for ALL of the following questions: $1 \times 5 = 5$

[transformer, polarization, intensity, interference, solenoid, AC generator]

16. A current carrying _____ is equivalent to a bar magnet.

17. The device which converts mechanical energy into electrical energy is _____.

18. The device that varies alternating voltage is called _____.

19. The phenomenon of light which confirms the transverse nature of light waves is _____.

20. During photoelectric emission, the number of photoelectrons emitted per second is directly proportional to the _____ of incident light.

PART - B

III. Answer the FIVE of the following questions:

$2 \times 5 = 10$

21. Sketch the electric field lines for (a) a positive point charge and (b) an electric dipole.

22. What are polar molecules? Give an example.

23. State and explain Ohm's law.

24. A current loop of magnetic dipole moment 10 Am^2 is freely suspended in a uniform magnetic field of $5 \times 10^{-3} \text{ T}$ and its plane is present in the direction of magnetic field. Calculate the torque that acts on the current loop.

25. Mention the two factors on which self-inductance of a solenoid depends.

26. What is displacement current? Write the mathematical form of Ampere- Maxwell law.

27. Define impact parameter. What is the angle of scattering for minimum impact parameter?

28. Give any two differences between intrinsic and extrinsic semiconductors.

PART - C

IV. Answer any FIVE of the following questions:

$3 \times 5 = 15$

29. Give Coulomb's law in vector form and explain the terms. Define SI unit of charge.

30. Mention any three important results regarding electrostatics of conductors.

31. What is Lorentz force? Write the expression representing this force and explain the terms.

32. State and explain Gauss's law in magnetism. Give its significance.

33. Derive the expression for the magnetic energy stored in a coil, $U = \frac{1}{2} LI^2$.

34. Give the Cartesian sign convention used to measure distances and heights in spherical mirrors.

35. Write any three properties of photon.

36. Calculate the binding energy of an Alpha(α) particle in MeV from the following data.

Mass of α -particle = 4.00260 u, Mass of neutron = 1.008662 u, Mass of proton = 1.007825 u

PART – D

V. Answer any **THREE** of the following questions:

5 × 3 = 15

37. Using Gauss's law, arrive at the expression for electric field at a point outside the charged spherical conducting shell. Also write the expression electric field at a point on the surface of the same charged spherical shell.
38. Obtain the expressions for equivalent emf and equivalent internal resistance of two cells of different emfs and different internal resistances connected in series.
39. Derive the expression for force per unit length on two infinitely long straight parallel conductors carrying currents and hence define 'ampere'.
40. (i) Two coherent waves of same amplitude and constant phase difference undergo interference. Obtain an expression for resultant displacement. (3)
(ii) Mention the two uses of polaroids. (2)
41. What is meant by rectification? Explain the working of a p-n junction diode as a full-wave rectifier, with the help of relevant circuit diagram and input-output waveforms.

VI. Answer any **TWO** of the following questions:

5 × 2 = 10

42. In a parallel plate capacitor with air between the plates, each plate has an area of 4 cm^2 and distance between the plates is 2 mm.
(a) Calculate the capacitance of the capacitor.
(b) If this capacitor is connected to a 100 V supply, then find the magnitude of charge on each plate and energy stored in the capacitor. Given: $\epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$.
43. A copper wire of length 50 cm and diameter 1.0 mm carries a current of 3.0 A. The number density of free electrons in copper is $8.5 \times 10^{28} \text{ m}^{-3}$. Calculate
(a) the current density,
(b) the drift velocity of free electrons and
(c) the time taken by the electrons to drift from one end of the wire to the other end.
[Given: $e = 1.6 \times 10^{-19} \text{ C}$]
44. A resistor, an inductor and a capacitor are connected in series with a 110V, 100Hz AC source. In the circuit, the voltage leads the current by 35° . If the resistance of the resistor is 10Ω and the sum of inductive reactance and capacitive reactance is 17Ω , then calculate the self-inductance of the inductor.
45. Find the effective focal length and effective power of combination of a convex lens of focal length 25cm in contact with a concave lens of focal length 20cm. Is the system a converging or a diverging lens? Ignore the thickness of the lenses.

Part-A.

I Pick the correct option among the given options for ALL of the following questions: $1 \times 15 = 15$

- 1) (b) (6) (c) (11) (a)
- 2) (a) (7) (c) (12) (c)
- 3) (d) (8) (c) (13) (b)
- 4) (a) (9) (d) (14) (b)
- 5) (b) (10) (b) (15) (c)

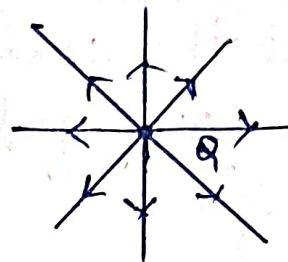
II Fill in the blanks by choosing appropriate answers given in the bracket for ALL of the following questions:-

- 16) Solenoid.
- 17) AC generator.
- 18) transformer.
- 19) Polarization.
- 20) Intensity.

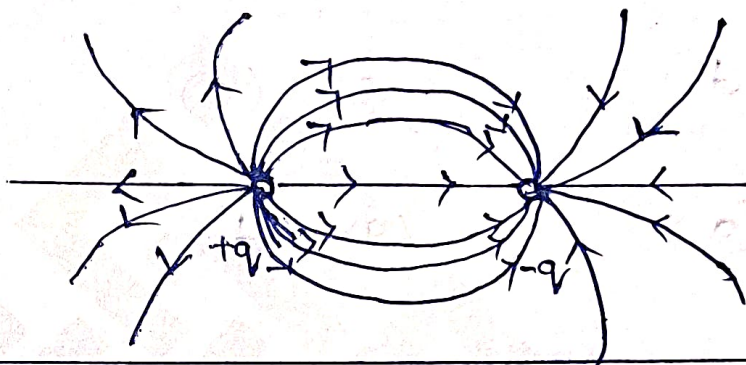
Part-B.

III Answer the five of the following questions: $2 \times 5 = 10$

21) (a) Positive point charge:-



(b) electric dipole:-



22) Polar molecules are molecules in which the centers of positive and negative charges do not coincide.

Examples: H_2O , HCl .

23) Statement: "The current in a metallic conductor is directly proportional to the potential difference applied across its ends, provided the temperature & other physical conditions of the conductor are kept constant."

Explanation: Ohm's law is given by $I \propto V \Rightarrow I = \frac{V}{R}$
 where, I = Current flowing in the conductor
 V = Potential difference applied across end of the conductor.

R = Resistance of the conductor.

24)

$$\tau = m B \sin \theta, \quad \theta = 90^\circ;$$

$$\tau = 10 \times 5 \times 10^{-3} \times \sin 90^\circ, \quad \sin 90^\circ = 1$$

$$\tau = 50 \times 10^{-3} \text{ N.m.}$$

$$\boxed{\tau = 0.05 \text{ N.m.}}$$

25)

$$L = \frac{\mu N^2 A}{l}$$

The Self-inductance of a Solenoid depends on the following key factors:-

* Number of turns (N).

* Cross-sectional area (A)

* length (l)

* permeability (μ).

26)

Displacement Current:-

It is the current due to time varying electric field.

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 (\vec{I}_c + \vec{I}_d) = \mu_0 \left(\frac{V}{R} + \epsilon_0 \frac{d\phi}{dt} \right)$$

$\vec{B}$ = magnetic field.

$d\vec{s}$ = infinitesimally small line element on the closed Path.

$\vec{I}_c$ = Conduction Current = V/R

$\vec{I}_d$ = Displacement Current = $\epsilon_0 \frac{d\phi}{dt}$.

27) Impact parameter:

It is defined as the perpendicular distance between the initial velocity vector of the α -particle and the central line of the nucleus.

$$b = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{2E} \cot\left(\frac{\theta}{2}\right)$$

$b \rightarrow 0$, $\theta \rightarrow 180^\circ$ for minimum impact parameter.

28)	Intrinsic Semiconductors	Extrinsic Semiconductors
1)	These are the pure Semiconductors.	These are the impure Semiconductors.
2)	Electron density is equal to the hole density.	Electron density is not equal to the hole density.
3)	Conductivity is low	Conductivity is high
4)	Conductivity depends on the temperature.	Conductivity depends on the temperature and also doping level.
5)	Ex:- Germanium (pure)	doped Germanium.

Part C.

W Answer any 5 of the following $3 \times 5 = 15$.

29)
$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

$\hat{r}$ = unit vector in the direction of $\vec{F}$.

q_1, q_2 = magnitude of the charges.

r = distance between the charges.

Define Coulomb or define unit of Charge:

1 Coulomb is the charge which repels another identical charge with a force of $9 \times 10^9 \text{ N}$, when placed at a distance of 1 meter in free space.

- 30)
- 1) The electric field inside a conductor in electrostatic equilibrium is zero. This means no electric field exists within the conducting material itself.
 - 2) Any excess charge placed on a conductor resides entirely on its surface. There is no excess charge inside the volume of the conductor.
 - 3) The electric field just outside a charged conductor is perpendicular to the conducting surface.
 - 4) All points on the surface of a conductor are at the same electric potential, making the surface an equipotential surface.
 - 5) Electrostatic Shielding occurs in conductors with cavity within a charged conductor is zero.
 - 6) The sharp points or edges on a conductor have higher surface charge.

31) Lorentz force:-

It is the resultant force experienced by the charged particle moving in a region where both magnetic and electric fields are present.

Lorentz force = Electric force + Magnetic force.

$$F = qE + B \times v \quad \text{or} \quad F = qE + Bv \sin \theta$$

q = magnitude of charge.

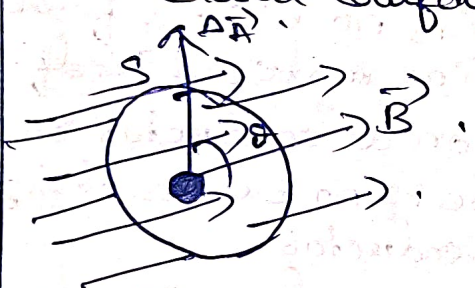
E = electric field.

B = magnetic field.

θ = angle between direction of B and v .

v = velocity of charged

32) Statement: "The net magnetic flux through any closed surface is zero."



Explanation:- In the given figure

S = closed surface.

$\vec{B}$ = uniform magnetic field

$\Delta \vec{A}$ = small area element

θ = angle between $\vec{B}$ & $\Delta \vec{A}$.

$$\Phi_B = \sum B \Delta A \cos \theta$$

* monopole magnet

$$\Phi_B = \sum \vec{B} \cdot \vec{\Delta A} = 0 \quad \text{does not exist.}$$

* Gauss's law is valid for any closed surface.

* It reveals that only external charges outside the closed surface do not contribute to the net electric flux through that surface.

33) Let dW be the work done in developing the current i in an inductor in time dt .

Let e = emf induced in an inductor.

The rate of work done is given by .

$$\frac{dW}{dt} = ei \quad \because P = \frac{W}{t} = \frac{eV}{t} = V i = ei.$$

$$\frac{dW}{dt} = L \frac{di}{dt} i \quad \because e = L \frac{di}{dt}.$$

$$dW = L i di.$$

Total work done in developing the current .

from $i=0$ to $i=I$.

$$W = \int_0^I dW.$$

$$W = \int_0^I L i di.$$

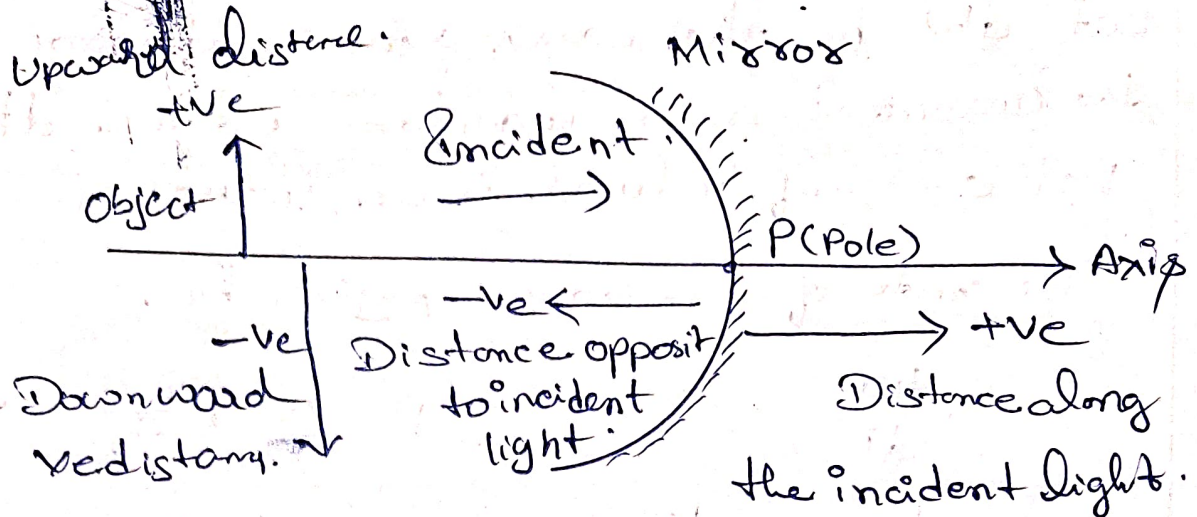
$$W = L \int_0^I i di.$$

$$W = \frac{1}{2} L I^2$$

$W = U$ = Stored magnetic potential energy.

$$U = \frac{1}{2} L I^2$$

34)



35)

- 1) Radiation consists of packets of energy called photons.
- 2) The rest mass of the photons is zero.
- 3) The photons are electrically neutral.
- 4) The photons travel with the speed of light.
- 5) The energy of each photon is given by $E = hf$
- 6) $p = mc$ or $p = \frac{h}{\lambda}$.

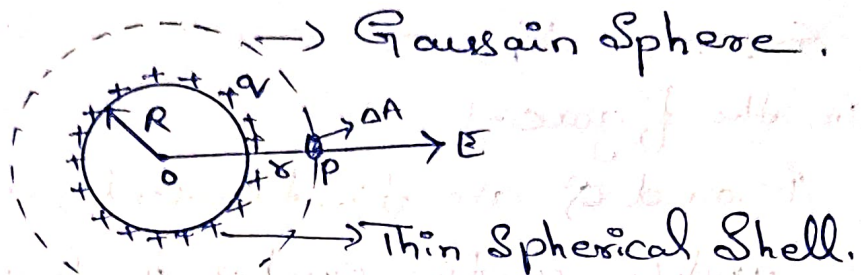
36.

α -particle: 2 proton, 2 neutrons.
 mass of α -particle: 4.00260 u.
 mass of 2 protons = $2 \times 1.007825 \text{ u} = 2.015650 \text{ u}$
 mass of 2 neutrons = $2 \times 1.008662 \text{ u} = 2.017324 \text{ u}$.
 Total mass of constituents: $2.015650 + 2.017324 = 4.032974 \text{ u}$.
 Mass defect $\Delta m = 4.032974 - 4.00260 = 0.030374 \text{ u}$
 Binding energy $= \Delta m \times 931.5 \text{ MeV}$
 $B.E. = 0.030374 \times 931.5 \text{ MeV} = 28.3 \text{ MeV}$
 $B.E. = 28.3 \text{ MeV}$

Part-D.

5 x 3 = 15.

Answer any Three of the following questions:-



In the figure:-

E = electric field at the point 'P'

ΔA = area element around the point 'P'

Q = charge in the thin spherical shell.

R = radius of the thin spherical shell.

r = radius of gaussian sphere.

The electric flux through the gaussian sphere is given by.

$$\phi = \sum E \Delta A \cos \theta$$

$$\phi = \sum \Delta A \cdot E \cos \theta \rightarrow (1)$$

Here, the angle θ between E and ΔA is 0° . $\therefore \cos 0^\circ = 1 \rightarrow (2)$

$\sum \Delta A = 4\pi r^2$ = area of the gaussian sphere. $\rightarrow (3)$

$$\phi = E \times 4\pi r^2 \rightarrow (4), \quad \phi = \frac{Q}{\epsilon_0} \rightarrow (5)$$

$$E \times 4\pi r^2 = \frac{Q}{\epsilon_0}$$

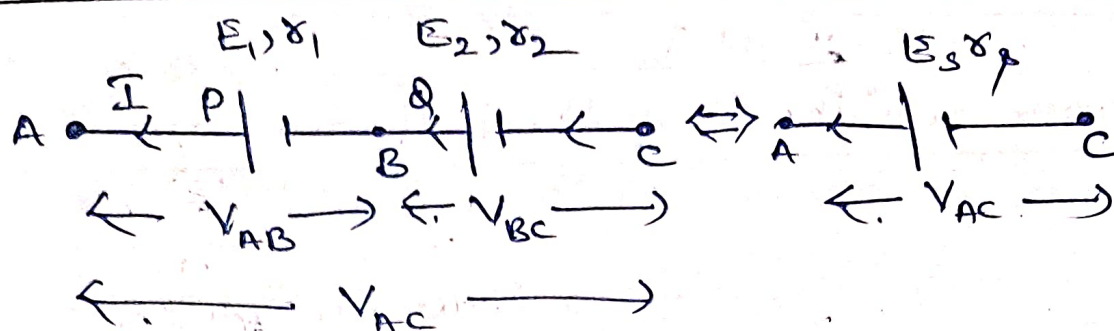
$$\boxed{E = \frac{1}{4\pi \epsilon_0} \frac{Q}{r^2}}$$

The direction of E is along OP produced.

$$\boxed{E = \frac{\sigma}{\epsilon_0}}$$

σ = Surface density ϵ_0 = permittivity of free space
is expression for electric field at a point on the surface of the same charged spherical shell.

38)



In the figure-1

P and Q are the two cells connected in series. E_1 and E_2 are the emf's of P and Q respectively. r_1 and r_2 are the internal resistance of P and Q respectively.

I is the current sent by the two cells.

V_{AB} , V_{BC} and V_{AC} = potential difference b/w the points A & B, B & C, & A & C respectively.

$$E = V + Ir$$

$$\Rightarrow V = E - Ir$$

$$P = V_{AB} = E_1 - Ir_1 \rightarrow (1)$$

$$Q = V_{BC} = E_2 - Ir_2 \rightarrow (2)$$

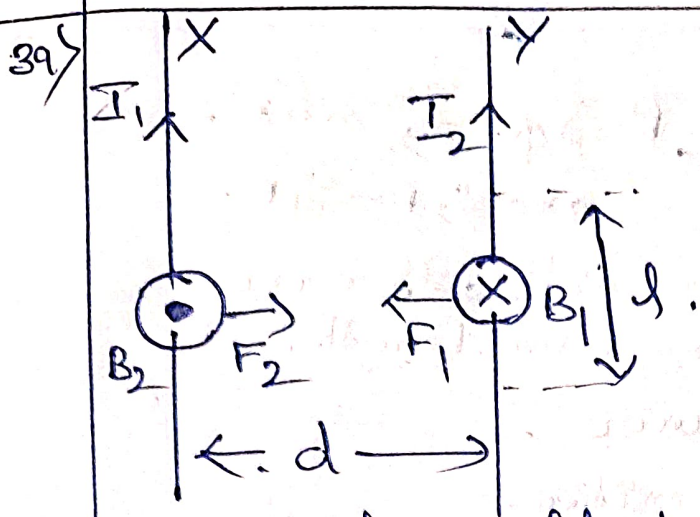
$$V_{AC} = V_{AB} + V_{BC} \rightarrow (3)$$

$$V_{AC} = (E_1 - Ir_1) + (E_2 - Ir_2) \rightarrow (4)$$

$$V_{AC} = (E_1 + E_2) - I(r_1 + r_2) \rightarrow (5)$$

$$V_{AC} = E_s - Ir_s \rightarrow (6)$$

$E_s = E_1 + E_2$
$r_s = r_1 + r_2$



On the figure.

X and Y = two infinitely long straight parallel conductors.
 I_1 and I_2 = current in X and Y respectively.

d = Perpendicular distance b/w the conductors.

Magnetic field due to the current I_1 is.

$$B_1 = \frac{\mu_0 I_1}{2\pi d} \rightarrow (1)$$

The direction of B_1 and B_2 we found using right hand grasp rule.

So the line segment l of conductor 'Y' is

$$F_1 = B_1 I_2 l \sin \theta$$

$$\therefore F = I l B \sin \theta$$

$$F_1 = B_1 I_2 l \rightarrow (2)$$

$$\theta = 90^\circ \therefore \sin 90^\circ = 1$$

$$\Rightarrow F_1 = \frac{\mu_0 I_1 I_2 l}{2\pi d} \quad \text{iii) } F_2 = \frac{\mu_0 I_1 I_2 l}{2\pi d}$$

The direction of F_1 & F_2 is attractive, According to Fleming's left hand rule.

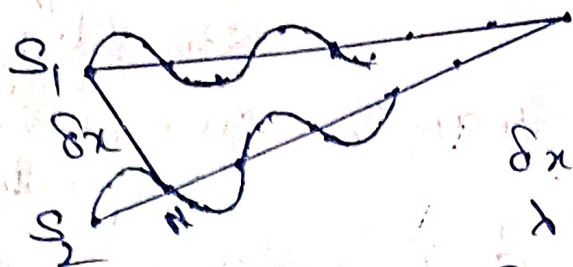
$$F = \frac{\mu_0 I_1 I_2 l}{2\pi d}$$

40)

$\delta\phi$ = phase difference

$$P \cdot \delta\phi = \frac{2\pi}{\lambda} \times \delta x$$

$$\delta x = S_2 P - S_1 P$$



δx = Path difference

λ = wave length.

S_1 & S_2 = Coherent Source.

P = point of Superposition.

SN = normal drawn to $S_1 P$.

let A = Amplitude, λ = wave length

$$Y_1 = A \sin(\omega t) \rightarrow \text{①}$$

$$Y_2 = A \sin(\omega t + \phi) \rightarrow \text{②}$$

$$Y = Y_1 + Y_2$$

$$Y = A \sin(\omega t) + A \sin(\omega t + \phi)$$

$$Y = A \left[2 \sin \left(\frac{\omega t + \omega t + \phi}{2} \right) \cdot \cos \left(\frac{\cancel{\omega t} - \cancel{\omega t} - \phi}{2} \right) \right]$$

$$Y = 2A \sin \left(\frac{2\omega t + \phi}{2} \right) \cdot \cos \left(-\frac{\phi}{2} \right)$$

$$Y = 2A \cos \left(\frac{\phi}{2} \right) \cdot \sin \left(\frac{2\omega t + \phi}{2} \right)$$

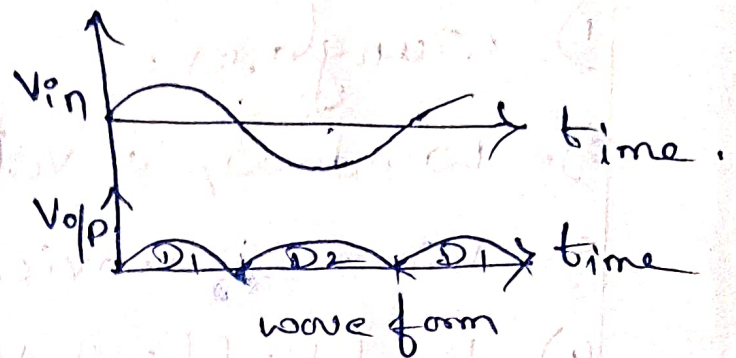
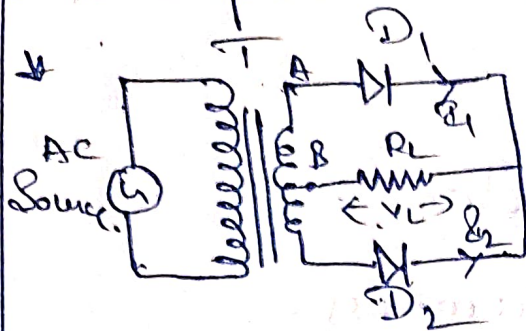
$$\boxed{Y = R \sin \left(\frac{2\omega t + \phi}{2} \right)}$$

$$\therefore R = 2A \cos \left(\frac{\phi}{2} \right)$$

Application of polaroids :- used in .

- 1) sunglasses.
- 2) windows of vehicles.
- 3) View 3D movie cameras.
- 4) Head lights of vehicles.
- 5) Produce & analyze the plane polarized light
- 6) windows of aeroplanes.
- 7) photographic cameras filters.
- 8) Improve colour contrast in old paintings.

41) → The process of conversion of a.c into d.c.



The circuit connection are made for full wave rectifier as shown in the circuit diagram.

The circuit contains ac. source, step down centre tapped transformer (T) two diodes (D_1 & D_2) and a load resistor (R_L).

During positive half cycle of input a.c the point A is positive with respect to the point B. Then D_1 is forward biased and D_2 is reverse biased. Hence current I_1 flows through R_L .

During negative half cycle of input a.c the point A is negative with respect to the point B. Then D_1 is reverse biased and D_2 is forward biased. Hence current I_2 flows through R_L .

Thus the diode allows both positive & negative half cycle of input a.c through R_L .

Hence the diode acts as full wave rectifier.

$$5 \times 2 = 10$$

Answer Any two of the following questions:-

42)

Given, $A = 4 \text{ cm}^2 = 4 \times 10^{-2} \text{ m}^2$

$d = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$

$C = ?$

$V = 100 \text{ V}$

$q = ?$

$\epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$

$U = ?$

$$C = \frac{\epsilon_0 A}{d} = \frac{8.854 \times 10^{-12} \times 4 \times 10^{-2}}{2 \times 10^{-3}}$$

$$C = \frac{35.416 \times 10^{-12} \times 10^{-2} \times 10^3}{2}$$

$$C = 17.708 \times 10^{-11} \text{ F}$$

$q = CV$

$q = 17.708 \times 10^{-11} \times 100$

$$q = 17.708 \times 10^{-9} \text{ C}$$

$$U = \frac{q^2}{2C} = \frac{17.708 \times 17.708 \times 10^{-9} \times 10^{-9}}{2 \times 17.708 \times 10^{-11}}$$

$U = 8.854 \times 10^{-9-9+11}$

$$U = 8.854 \times 10^{-7} \text{ J}$$

43)

$$L = 50 \text{ cm} = 0.5 \text{ m}$$

$$D = 1 \text{ mm} = 0.001 \text{ m}$$

$$r = 0.0005 \text{ m}$$

$$I = 3 \text{ A}, \quad e = 1.6 \times 10^{-19} \text{ C}$$

$$n = 8.5 \times 10^{28} \text{ m}^{-3}$$

$$a) \quad J = ?$$

$$b) \quad V_d = ?$$

$$c) \quad t = ?$$

$$a) \quad J = \frac{I}{A} = \frac{3}{\pi r^2} = \frac{3}{3.142 \times (0.0005)^2}$$

$$J = \frac{3}{7.855 \times 10^{-7}} = \frac{3}{7.8 \times 10^{-7}} = 0.3819 \times 10^7$$

$$\boxed{J = 3.8192 \times 10^6 \text{ A/m}^2}$$

$$b) \quad V_d = \frac{I}{n A e} = \frac{3}{8.5 \times 10^{28} \times 7.855 \times 10^{-7} \times 1.6 \times 10^{-19}}$$

$$\boxed{V_d = 0.2808 \times 10^{-3} \text{ m/s}}$$

$$c) \quad V_d = \frac{l}{t} \Rightarrow t = \frac{l}{V_d} = \frac{0.5}{0.2808 \times 10^{-3}}$$

$$\boxed{t = 1.7806 \times 10^3 \text{ s}}$$

44)

$$V = 110 \text{ V}$$

$$f = 100 \text{ Hz}$$

$$\phi = 35^\circ$$

$$R = 10 \Omega$$

$$(X_L + X_C) = 17 \Omega$$

$$L = ?$$

$$\tan \phi = \frac{(X_L - X_C)}{R}$$

$$(X_L - X_C) = R \cdot \tan \phi$$

$$(X_L - X_C) = 10 \times \tan 35^\circ$$

$$(X_L - X_C) = 10 \times 0.7002$$

$$(X_L - X_C) = 7.002 \rightarrow \textcircled{1}$$

$$X_L + X_C = 17 \rightarrow \textcircled{2} \text{ Given.}$$

$$\Rightarrow X_L + X_C = 17$$

$$X_L - X_C = 7$$

$$2X_L = 24$$

$$\boxed{X_L = 12 \Omega}$$

$$-X_C = 17 - 12$$

$$\boxed{X_C = 5 \Omega}$$

$$X_L = 2\pi f L$$

$$L = \frac{X_L}{2\pi f}$$

$$L = \frac{12}{2 \times 3.142 \times 100}$$

$$L = \frac{12}{6284} = 0.01909 \text{ H.}$$

$$\therefore \boxed{L = 0.01909 \text{ H.}}$$

45) $f_1 = 25 \text{ cm} = 0.25 \text{ m}$, $P_1 = ?$
 $f_2 = -20 \text{ cm} = -0.20 \text{ m}$, $P_2 = ?$

$F = ?$, $P = ?$

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{F} = \frac{f_1 + f_2}{f_1 \cdot f_2}$$

$$F = \frac{f_1 \times f_2}{f_1 + f_2}$$

$$F = \frac{(0.25) \times (-0.20)}{[0.25 - 0.20]}$$

$$\boxed{F = -1 \text{ m}} \text{ or } \boxed{F = -100 \text{ cm}}$$

$$P_1 = \frac{1}{f_1} = \frac{1}{0.25} = 4 \text{ D} \quad \therefore P_1 = 4 \text{ D}$$

$$P_2 = \frac{1}{f_2} = \frac{1}{-0.2} = -5 \text{ D} \quad \therefore P_2 = -5 \text{ D}$$

$$P = \frac{1}{F} = \frac{1}{-1} = -1 \text{ D} \quad \therefore \boxed{P = -1 \text{ D}}$$

'The system is diverging lens'